

11 A bowling ball of mass 4.50 kg falls a short distance vertically into a tray of sand.

The bowling ball decelerates uniformly in the sand.

The deceleration of the bowling ball when it is in contact with the sand is $8.00 \times 10^2 \text{ m s}^{-2}$.

What is the magnitude of the force exerted on the bowling ball by the sand as it decelerates?

A 44.1 N

B 3560 N

C 3600 N

D 3640 N